SQL fundamentals: Select & filtering.

SELECTING & SORTING

1. Retrieve all the columns from the employee table

```
SELECT * FROM employees;
```

ID	First_Name	Last_Name	Department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing
5	Eva	Naidoo	IT.

2. Find all unique departments.

```
SELECT DISTINCT department
FROM employees;
```

Department
IT
HR
Finance
Marketing

3. Retrieve first and last names ordered by salary descending

```
SELECT first_name, last name
FROM employees
ORDER BY salary DESC;
```

First_Name	Last_Name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees.

```
SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3
```

ID	First_name	Last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	7000
3	Cathy	Zulu	65 000

FILTERING with WHERE

5. find employees in the IT department

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';
```

ID	First_name	Last_name	Department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Find employees in Finance with salary > 6000

```
SELECT id, first_name, last_name, salary
FROM employees
WHERE department = 'Finance'
AND salary > 6000;
```

ID	First_name	Last_name	Salary
3	Cathy	Zulu	65 000

07. Find employees in HR or Marketing

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department 'HR'
OR department = 'Marketing';
```

ID	First_Name	Last_name	Department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find employees not in HR

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department <> 'IT';
```

ID	First_Name	Last_Name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Find employees in IT, HR or Finance using N

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

ID	First_Name	Last_name	Department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 6000 & city Johannesburg

```
SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';
```

ID	First name	Last-name	Department
1	Alice	Green	IT
5	Eva	Naidoo	IT